A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE
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Intelligent Tweet Analyzer for Disaster Management

Real-time Social Media Intelligence for Emergency Response

BE FINAL YEAR PROJECT · CSE (AI & ML)

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Abstract

Social media platforms like Twitter (now X) serve as vital real-time channels during disasters — floods, earthquakes, and pandemics. This project leverages Twitter data and **Natural Language Processing (NLP)** + **Machine Learning (ML)** to automate tweet analysis, classify emergencies, and support rapid decision-making during crises.

The system integrates **geolocation tagging**, **sentiment analysis**, and **urgency scoring** to enhance situational awareness and help response teams prioritize resources efficiently.

NLP · Machine Learning · Sentiment Analysis · Geolocation · Real-Time Response · Emergency Classification

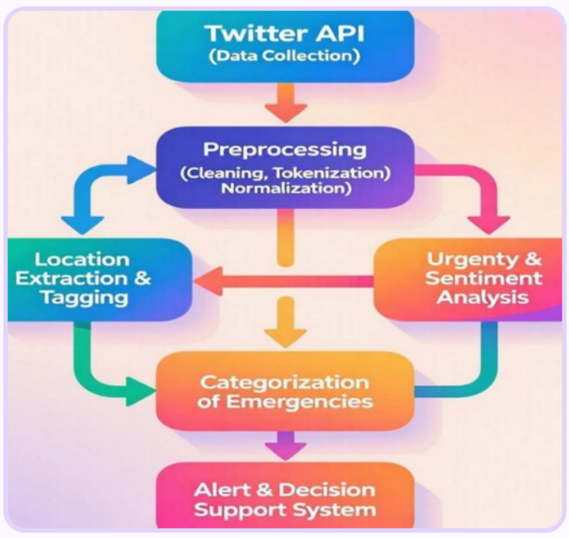
Key Applications

1. Flood Response
2. Wildfire Alerts
3. Medical Emergencies
4. Earthquake Relief
5. Hurricane Monitoring
6. First Responder Aid

Objectives

- Develop an automated system to collect and preprocess real-time disaster-related tweets from Twitter API
- Extract and tag geographical locations from tweet content to enable geospatial prioritization of rescue efforts
- Perform urgency scoring and sentiment analysis to detect distress and identify high-priority situations
- Classify tweets into actionable emergency categories to streamline decision-making for first responders
- Provide an interactive real-time dashboard with heatmaps, analytics, and visual insights
- Integrate with existing disaster management frameworks to improve resource allocation and response times

System Architecture & Workflow



- **Twitter API** — Real-time tweet collection using disaster-related keywords & location filters
- **Preprocessing** — Cleaning (remove URLs, stopwords, emojis), tokenization & normalization
- **Feature Extraction** — TF-IDF vectorization and word embeddings for numerical representation
- **AI Classification** — Google Gemini LLM & BiLSTM-CNN classify into 5 emergency categories
- **Geolocation & NER** — Named entity recognition extracts location, victim count & resources
- **Alert Engine** — Urgency scoring triggers automated alerts to first responders & NGOs
- **Dashboard** — Live incident feed, heatmaps, analytics, and visual insights

TECHNOLOGIES USED

BACKEND
1. Python 3.10+
2. FastAPI
3. MongoDB
4. Docker

FRONTEND
1. React.js
2. Google Maps API
3. Chart.js
4. Twitter API

AI / NLP
1. Google Gemini API
2. TensorFlow
3. Hugging Face

ML LIBRARIES
1. Scikit-learn
2. PyTorch
3. NLTK / SpaCy

Methodology

- **Data Collection:** Real-time tweets fetched via Twitter API using disaster-related keywords and location filters
- **Preprocessing:** Text cleaning — removal of URLs, stopwords, emojis, hashtags; tokenization and normalization
- **Feature Extraction:** TF-IDF vectorization and word embeddings for numerical representation of tweet content
- **Classification:** Google Gemini LLM and BiLSTM-CNN models classify tweets into 5 actionable emergency categories
- **Geolocation & NER:** Named entity recognition extracts location, victim count, and resource mentions from tweets
- **Alert Generation:** Urgency scoring triggers automated alerts to first responders and NGOs via dashboard

Output

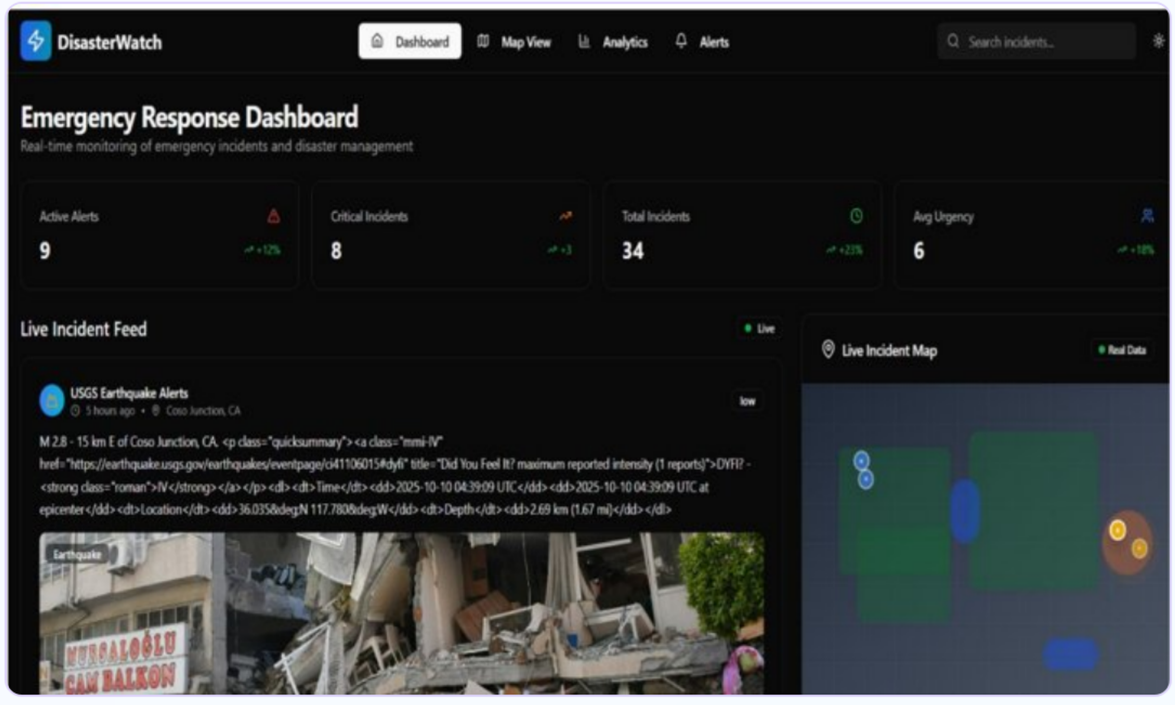


Fig: Emergency Response Dashboard — live incident feed, active alerts, critical incidents count, and real-time incident map.

The analytics dashboard shows a consistent detection of incidents with a high resolution rate (above 90%), confirming the robustness of the NLP-based preprocessing and classification pipeline in handling informal, unstructured textual data.

Conclusion

The **Intelligent Tweet Analyzer for Disaster Management** bridges the gap between social media intelligence and operational disaster response. By combining NLP, deep learning, and real-time geolocation, the system automates tweet classification, urgency detection, and alert generation — enabling faster, targeted emergency responses.

The framework handles noisy, unstructured data and integrates seamlessly with disaster management workflows, improving situational awareness, resource allocation efficiency, and ultimately contributing to saving lives during critical events.

Future Scope

Multilingual Support
Regional language support (Hindi, Tamil) for broader coverage

Mobile App
On-ground disaster management personnel mobile application

Multimodal Analysis
Images & videos from disaster tweets for richer context

SMS/Push Alerts
Real-time notification alerting for field responders

References

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